

Introduction

Vaut-il mieux inclure les références aux chapitres dans le texte, ou faire une phrase à la fin de chaque paragraphe pour énoncé le contenu de chaque chapitre ? (cf Mathias' thesis)

Faut-il ajouter une illustration du plan de la thèse ?

The modern cosmology is based on the Λ -CDM theoretical model, described in Chapter 1, which suggests the existence of an exponentially expanding phase in the very early Universe. This phase, called Inflation, occupies a fundamental position in the model, as it is inseparable from our understanding of our Universe. However, no observational evidence has been found in order to justify this hypothesis. Then, the goal of the majority of the current cosmology experiments is to find such proof for Inflation. The main focus is the quest for primordial polarisation B-modes patterns in the Cosmic Microwave Background, detailed in Chapter 2, which can be caused by gravitational waves expected theoretically by the Inflation theory. Today, only an upper limit for the amplitude of this pattern can be set from current observation.

This absence of detection comes from the multiple difficulties associated with this observation. First, the expected amplitude of primordial B-modes is tiny, much smaller than former experiment sensitivity. Second, this signal is contaminated by various phenomena: astrophysical foregrounds, gravitational lensing, or atmospheric emissions. The next generation of experiments needs to address these issues by guaranteeing a tight control of systematic effects while allowing separation between primordial B-modes and their contamination. These are the reasons behind the QUBIC experiment, presented in Chapter 3, innovative design, called Bolometric Interferometry, which combines the high sensitivity of bolometers with unique features of interferometry. In particular, we can also increase our spectral resolution by splitting our physical bandwidth using spectral imaging during the data

analysis stage, which allows an improved component separation capability. A focus will be given to spectral imaging in Chapter 4, as it occupies a central place within this thesis. Additionally, interferometry will be a great addition for systematic control, as it allows for self-calibration, a method widely used in radio interferometry. Preliminary work toward this technique has been performed on QUBIC's calibration source and, in particular on its associated GPS, which can be found in Chapter 5. Chapter 6 will be dedicated to the software part of this experiment, as its development has been a large fraction of my time during this PhD.

The development of map-making methods relying on spectral imaging has been the main goal of this thesis. The first algorithm, called Frequency Map-Making and presented in Chapter 7, reconstructs frequency maps from data, with the particularity that we can reconstruct multiple maps within each physical band thanks to spectral imaging. A second and improved algorithm, named Component Map-Making, is introduced in Chapter 8 and aims at reconstructing directly astrophysical component maps, meaning that we are performing map-making and component separation during the same process, which is only possible with spectral imaging capability. Chapter 9 will develop the realism of data simulation associated with these two algorithms, which is a central part of the simulation software. It includes the effect of various parameters, an explanation of the treatment of convolution, and the addition of external data, alongside hyperparameter optimisation. We will introduce in Chapter 10 an alternative for the two algorithms, which are based on forward modelling, by proposing an approximation of the inverse map-making problem using neural networks.

While all the previous element are improvement of existing map-making or astrophysical component separation methods, possible with spectral imaging, we wanted to use them on another main issue for ground-based experiments : atmospheric emissions. The goal of the Chapter 11 is to provide details of this work, going from developing of simulation for the atmosphere to mitigating its contribution in data by using the Component Map-Making algorithm, and comparing these results to the standard filtering method. The final chapter of this thesis, the Chapter 12, will conclude with Forecasts for QUBIC instrument. It will include comparison between different sky models and instrumental configurations, as well as between the various algorithms presented through this manuscript.